

Functions of Random Variables

CSU, Chico Math 350

2026-03-13

outline

Functions of RVs (Ch 3.4)

Independent RVs (Ch 3.5.1)

References

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Concept

There are many reasons why we might be more interested in looking at the distribution of a **function** of a random variable X than the actual variable X .

One example would be that we're interested in the absolute distance the random variable is away from its mean:

$g(x) = |X - \mu|$, or we want to know the total gain or loss in a stock portfolio by adding up all the sum of the daily results.

The PMF of $g(X)$ can be computed as follows. For each y , the probability that $g(X) = y$ is given by $\sum p(x)$ where the sum is over all values of x such that $g(x) = y$.

Example 1, Theoretical

Let $X = -2, -1, 0, 1, 2$, all equally likely, and $g(x) = X^2$. Find the PMF of $y = g(x)$, as well as $E(Y)$.

x	$p(x)$	$g(x) = X^2$
-2	0.2	4
-1	0.2	1
0	0.2	0
1	0.2	1
2	0.2	4

Notice that the same value for $y = g(x)$ can occur from multiple values of x . We simplify the PMF of Y to only have one row per unique value of Y , and use this same table to calculate $E(Y)$:

$Y = g(x)$	$p(y)$	$y * p(y)$
0	0.2	0
1	0.4	0.4
4	0.4	1.6

Therefore, $E(Y) = \sum y * p(y) = 0 + 0.4 + 1.6 = 2$

Example 1, Simulated

Confirm your results using simulation.

```
X <- c(-2, -1, 0, 1, 2)
sample.x <- sample(X, 10000, replace=TRUE) # sample from x
y <- sample.x^2 #y = g(x)
proportions(table(y)) #pmf
## y
##      0      1      4
## 0.1967 0.3965 0.4068

mean(y) # expected value
## [1] 2.0237
```

Note that no "prob" vector is needed here since all values of X have equal probability to occur.

Exercise 1

Using the random variable X from the previous example, find the PMF and expected value of $Y = 2X + 1$

STEPS

1. Determine the PMF (what would the table look like?)
2. Calculate the theoretical expected value. Write your answer in a full sentence in context of the problem.
3. Confirm using simulation.

x	$p(x)$	$g(x) = 2X + 1$

Exercise 2

John travels to work five days a week. We will use X_1 to represent his travel time on Monday, X_2 to represent his travel time on Tuesday, and so on.

STEPS

1. Write an equation using $X_1, \dots, X_5$ that represent John's total travel time for the week, denoted by W .
2. It takes John an average of 18 minutes each day to commute to work. What would you expect his average commute time to be for the week? Explain how you got this answer.
3. What was a major assumption that we had to make to figure out this example?

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Concept

We say that two random variables are independent if the outcome of X does not give probabilistic information about the outcome of Y and vice versa.

- * Give an example of 2 variables that you think are independent.
- * Give an example of 2 variables that you think are **not** independent.

Theorem 3.8: Rules of Expectation

For random variables X and Y , and constants a , b , and c :

$$E[aX + bY] = aE[X] + bE[Y] \quad \text{and} \quad E[c] = c$$

Refer back to the commute time example. We intuitively reasoned that the expectation of the total time is equal to the sum of the expected individual times.

This theorem generalizes and formalizes that statement to say that the **expectation of a sum of random variables is always the sum of the expectation for each random variable.**

Thm. 3.8 Example

1. Find $E(2X + 5)$ if $E(X) = 4$

$$E(2X + 5) = E(2X) + E(5)$$

$$= 2 * E(X) + 5$$

$$= 2(4) + 5$$

$$= 13$$

2. Find $E(2X + 5Y)$ if $E(X) = 4$ and $E(Y) = -2$

$$E(2X + 5Y) = E(2X) + E(5Y)$$

$$= 2 * E(X) + 5E(Y)$$

$$= 2(4) + 5(-2)$$

$$= -2$$

Thm. 3.8 Exercise

1. Let $E(X) = 2$. Find $E(3X - 1)$
2. Find $E(2)$
3. Find $E(2X - 3Y)$ when $E(X) = 4$ and $E(Y) = 1$

Theorem 3.9: Alternate Variance Calculation

Using the rules we have seen above, recall another approach to variance we saw earlier:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

x	$p(x)$	$x * p(x)$	x^2	$x^2 * p(x)$
-2	0.2	-0.4	4	0.8
-1	0.2	-0.2	1	0.2
0	0.2	0	0	0
1	0.2	0.2	1	0.2
2	0.2	0.4	4	0.8

$$E(X) = \sum(x * p(x)) = 0$$

$$E(X^2) = \sum(x^2 * p(x)) = 2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 2 - 0^2 = 2$$

Confirming via Simulation

```
X <- c(-2, -1, 0, 1, 2)
sample.x <- sample(X, 10000, replace=TRUE) # sample from x
mean(sample.x)
## [1] -0.0143

var(sample.x)
## [1] 2.014297
```

Thm. 3.9 Exercise

Let X be a random variable with values $x = 0, 1, 2$ and $p(x) = 0.1, 0.5, 0.4$. Find $E(X)$ and $Var(X)$.

STEPS

1. Determine the PMF (what would the table look like?) Note that the values are not equally likely.
2. Calculate $E(X)$ and $E(X^2)$, and use these to find $Var(X)$
3. Confirm using simulation.

x	$p(x)$	$x * p(x)$	x^2	$x^2 * p(x)$

Theorem 3.10: Rules of Variance

1. Let X be a random variable and c be a constant. Then:

$$\text{Var}(cX) = c^2 \text{Var}(X) \quad \text{and} \quad \text{Var}(c) = 0$$

2. Let X and Y be independent random variables. Then:

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Thm. 3.10 Example

Suppose that three random variables X_1, X_2, X_3 form a random sample from a distribution for which the mean is 5 and the variance is 3. Determine the value of $E(2X_1 - 3X_2 + X_3 - 4)$ and $Var(2X_1 - 3X_2 + X_3 - 4)$.

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$$\begin{aligned} & E(2X_1 - 3X_2 + X_3 - 4) \\ &= 2E(X_1) - 3 * E(X_2) + E(X_3) - 4 \\ &= -4 \end{aligned}$$

$$\begin{aligned} & Var(2X_1 - 3X_2 + X_3 - 4) \\ &= 4 * Var(X_1) + 9 * Var(X_2) + Var(X_3) \\ &= 4 * 3 + 9 * 3 + 3 \\ &= 42 \end{aligned}$$

Thm. 3.10 Exercise

In a marksmanship competition, each contestant takes ten shots with each of two different handguns. Final scores are computed by taking a weighted average of 4 times the number of bull-eyes made with the first gun plus 6 times the number gotten with the second gun.

If Bertha has a 30% chance of hitting the bull's-eye with each shot from the first gun and a 40% chance with each shot from the second gun, what is the theoretical mean and standard deviation of her score?

Additional info: Variance for the first gun is 2.1, variance for the second gun is 2.4.

STEPS

1. Define the random variables and function to create her final score.
2. Calculate $E(X)$ and $E(X^2)$, and use these to find $\text{Var}(X)$

Thm. 3.10 Exercise

Confirm with simulation.

TIPS:

- * Defining separate samples for each gun will be useful here, with a way to tell bull's-eyes and non-bull's-eyes apart.
- * Remember, the guns have different probabilities of hitting the bull's-eye.

This is really the key of why we practice simulation. Sometimes you don't have the ability to calculate the theoretical values, but you can simulate the experiment to get an estimate of the values.

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.